

ALI RAZA KHAN

AI Engineer — AI/ML · Data Science · C++

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PROFESSIONAL SUMMARY

Software Engineer with production **C++** experience (**Unreal Engine**) and hands-on **ML** and **Data Science** engineering. Fine-tuning transformer models (**DistilBERT**, **ViT**), building demand forecasting pipelines (**XGBoost**, **LightGBM**), and deploying end-to-end with **FastAPI**, **NodeJS** and **React**. Bringing real-time systems and performance-engineering discipline from game development into applied **AI**, **Data Science**, and **Computer Vision**.

TECHNICAL SKILLS

Languages: Python, C++, C#, SQL, JavaScript

AI / ML: PyTorch, Hugging Face Transformers, model fine-tuning (DistilBERT, ViT), Scikit-learn, OpenCV, NumPy, Pandas.

Backend & Web: FastAPI, REST API development, NodeJS, React (Vite), HTML, CSS.

Real-Time & Game: Unreal Engine (C++ & Blueprints), Unity (C#), AI Behavior Trees, multithreading, game physics

Tools: Git, VS Code, Jupyter, NVIDIA Omniverse, Windows/Linux dev environments

AI / ML PROJECTS

Walmart Demand Forecasting Pipeline | [Repo Link](#)

- Built an end-to-end demand forecasting pipeline on **421,570 rows** of Walmart weekly sales data, merged three data sources, engineered lag features (**1/4/52-week**), rolling means, and calendar decomposition across 45 stores and 81 departments.
- Benchmarked XGBoost vs LightGBM with early stopping on a chronological train-test split; achieved **MAE \$1,374** and **RMSE \$3,028**. Feature importance revealed **Sales_Lag_52** (same-week-last-year) as the dominant signal. .

The Daily AI — News Topic Classifier | [Repo Link](#)

- Fine-tuned **DistilBERT** on the AG News dataset for multi-class news classification, handling the full pipeline: dataset preparation, training, evaluation, and export.
- Deployed the model behind a FastAPI REST backend with a React (Vite) newspaper-themed frontend for live inference.

AI vs. Real Image Detector | [Repo Link](#)

- Fine-tuned a **Vision Transformer (ViT)** to classify AI-generated vs. real images; built evaluation workflow and inference API.
- Developed a drag-and-drop web interface (React) for real-time image classification with confidence scores.

Deepfake Detection System | Final Year Project - [Repo Link](#)

- Built an end-to-end deepfake detection pipeline (feature extraction, preprocessing, model training and evaluation) with Python, Scikit-learn, NumPy, and Pandas.
- Deployed behind a web interface enabling users to upload videos and receive real/fake classification.

PROFESSIONAL EXPERIENCE

Associate Software Engineer (Unreal Engine) | Devsinc

Jun 2025 – Present

- Design and implement real-time gameplay systems in C++ and Blueprints, combat mechanics, modular weapon systems, and interactive features, within a professional product team.
- Build AI behavior pipelines (Behavior Trees) for NPCs; apply component-based architecture, code review, and version-control best practices.

Game Development Intern (Unreal Engine) | Dreamcast Interactive

May 2025 – Jun 2025

- Developed gameplay prototypes in C++/Blueprints; explored multiplayer architecture including client-server networking, replication, and RPCs.

Game Development Intern (Unity) | Mindstorm Studios

Jun 2024 – Sep 2024

- Developed a choice-based visual novel in Unity (C#) from concept to completion, including custom UI animation systems and structured narrative data design.

Web Development | Lahore University of Management Sciences (LUMS)

Sep 2020 – Nov 2020

- Built CyberKhel.com, a gaming/entertainment website (WordPress, HTML, CSS).

EDUCATION & CERTIFICATIONS

BS Computer Science | Lahore Garrison University

2021 – 2025

Certifications: NVIDIA Omniverse Extensions Development · Rookie Game Jam 2024 (Mindstorm Studios) · Web Development (LUMS)